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Q1. Arrays, Array Applications and Structures**CODE**

```
#include <stdio.h>
#include <string.h>
int main() {
    char str[1000]="aabbccdd";
    int freq[256]={0};
    for(int i=0;str[i]!='\0';i++){
        freq[(unsigned char)str[i]]++;
    }
    for(int i=0;str[i]!='\0';i++){
        if(freq[(unsigned char)str[i]]==1){
            printf("%c",str[i]);
            return 0;
        }
    }
    printf("-1");
    return 0;
}
```

INPUT

aabbccdd

OUTPUT

(no output)

Q2. Arrays, Array Applications and Structures**CODE**

```
#include <stdio.h>
int main() {
    int n,target;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    scanf("%d",&target);
    int low=0,high=n-1;
    while(low≤high){
        int mid=low+(high-low)/2;
        if (arr[mid]==target){
            printf("%d",mid);
            return 0;
        }
        if (arr[low]≤arr[mid]){
            if(target≥arr[low] && target<arr[mid])
                high=mid-1;
            else
                low=mid+1;
        }
        else{
            if (target>arr[mid] && target≤arr[high])
                low=mid+1;
            else
                high=mid-1;
        }
    }
    printf("-1");
    return 0;
}
```

INPUT

```
7
4 5 6 7 0 1 2
6
```

OUTPUT

(no output)

Q3. Arrays, Array Applications and Structures**CODE**

```
#include <stdio.h>
int main() {
    int n;
    scanf("%d",&n);
    int intervals[n][2];
    for (int i=0;i<n;i++){
        scanf("%d %d",&intervals[i][0],&intervals[i][1]);
    }
    for(int i=0;i<n-1;i++){
        for(int j=i+1;j<n;j++){
            if(intervals[i][0]>intervals[j][0]){
                int t1=intervals[i][0];
                int t2=intervals[i][1];
                intervals[i][0]=intervals[j][0];
                intervals[i][1]=intervals[j][1];
                intervals[j][0]=t1;
                intervals[j][1]=t2;
            }
        }
    }
    int start=intervals[0][0];
    int end=intervals[0][1];
    for(int i=1;i<n;i++){
        if(intervals[i][0]≤end){
            end=intervals[i][1];
        }
        else{
            printf("%d %d\n",start,end);
            start=intervals[i][0];
            end=intervals[i][1];
        }
    }
    printf("%d %d",start,end);
    return 0;
}
```

INPUT

```
4
1 3
2 6
8 10
15 18
```

OUTPUT

(no output)

Q4. Arrays, Array Applications and Structures

CODE

```
#include <stdio.h>
#include <string.h>
int main() {
    char str[1000]="tree";
    int freq[256]={0};
    for(int i=0;str[i]!='\0';i++){
        freq[(unsigned char)str[i]]++;
    }
    while (1){
        int max=0;
        int ch=-1;
        for(int i=0;i<256;i++){
            if (freq[i]>max){
                max=freq[i];
                ch=i;
            }
        }
        if(max==0)
            break;
        for(int i=0;i<max;i++){
            printf("%c",ch);
            freq[ch]=0;
        }
        return 0;
    }
}
```

INPUT

tree

OUTPUT

(no output)

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Q1. Arrays, Array Applications and Structures

— Not Submitted —

Q2. Arrays, Array Applications and Structures

— Not Submitted —

Q3. Arrays, Array Applications and Structures

— Not Submitted —

Q4. Arrays, Array Applications and Structures

— Not Submitted —